

Chapter 17: Complex Numbers, Powers & De Moivre's Formula

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering

Course: Applied Ordinary Differential Equations (ENGR 213)

Textbook:

Advanced Engineering Mathematics

(7th Edition) by Dennis G. Zill — Chapter 17 (§17.1, §17.2)

1. Executive Overview & First-Principles Philosophy

In elementary algebra, the real number line $\mathbb{R}$ suffices for measurement, but it fails to be **algebraically closed**: simple polynomial equations like $x^2 + 1 = 0$ or $x^2 - 4x + 13 = 0$ have no real solutions.

In applied engineering mathematics, differential equations describing physical reality frequently lead to characteristic equations with negative discriminants:

- **Mechanical Vibrations:** The equation $mx'' + cx' + kx = 0$ for underdamped systems yields characteristic roots $r = -\frac{c}{2m} \pm i\sqrt{\frac{k}{m} - \frac{c^2}{4m^2}}$.
- **AC Electrical Circuits:** Alternating currents and voltages vary sinusoidally as $V(t) = V_0 \cos(\omega t + \phi)$. Expressing impedances and phasors in the complex plane converts complex differential calculus into straightforward complex algebra ($V = IZ$).
- **Fluid Dynamics & Aerodynamics:** Potential flow around airfoils is analyzed using complex velocity potentials $F(z) = \phi(x, y) + i\psi(x, y)$ through conformal mapping.

The **Complex Number System** $\mathbb{C}$ expands the 1D real number line into a 2D geometric plane (the Argand plane):

- The imaginary unit i satisfies $i^2 = -1$.
- Geometrically, multiplying any number by i corresponds to a **counterclockwise rotation of 90° ($\pi/2$ radians)** in the plane without altering its length.
- Euler's identity $e^{i\theta} = \cos \theta + i \sin \theta$ establishes the profound connection between exponential growth and circular trigonometry.
- De Moivre's Formula empowers engineers to raise complex quantities to any integer power and extract all n distinct roots with elegant geometric symmetry.

2. Mathematical Framework & Complex Algebra (Zill §17.1)

2.1 The Rectangular Representation & Algebraic Operations

A. Definition of a Complex Number

A **complex number** z is an ordered pair of real numbers (x, y) , conventionally written in Cartesian rectangular form as:

$$z = x + iy$$

where:

- $x = \text{Re}(z)$ is the **real part** of z .
- $y = \text{Im}(z)$ is the **imaginary part** of z (note: $\text{Im}(z)$ is a real number; it is the coefficient of i).
- i is the **imaginary unit** defined by $i^2 = -1$.
- If $y = 0$, $z = x$ is a purely real number ($\mathbb{R} \subset \mathbb{C}$).
- If $x = 0$ and $y \neq 0$, $z = iy$ is called a **pure imaginary number**.

B. Cyclic Powers of i

Powers of i repeat every four integer steps:

$$i^0 = 1, \quad i^1 = i, \quad i^2 = -1, \quad i^3 = i^2 \cdot i = -i, \quad i^4 = (i^2)^2 = 1$$

For any integer $n \in \mathbb{Z}$, divide n by 4 with remainder $r \in 0, 1, 2, 3$:

$$i^n = i^{4q+r} = (i^4)^q \cdot i^r = (1)^q \cdot i^r = i^r$$

- Example

$$\therefore i^{75} = i^{4(18)+3} = i^3 = -i.$$

- Example

$$\therefore i^{-1} = \frac{1}{i} = \frac{i}{i^2} = \frac{i}{-1} = -i.$$

C. Algebraic Operations in Rectangular Form

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$:

1. **Equality:**

$$z_1 = z_2 \iff x_1 = x_2 \text{ and } y_1 = y_2$$

Crucial Engineering Rule

: A single complex equation $A + iB = C + iD$ is always equivalent to **two independent real equations**: $A = C$ and $B = D$.

2. **Addition and Subtraction:**

$$z_1 \pm z_2 = (x_1 \pm x_2) + i(y_1 \pm y_2)$$

3. **Multiplication:** Applying the standard distributive law and using $i^2 = -1$:

$$z_1 z_2 = (x_1 + iy_1)(x_2 + iy_2) = x_1 x_2 + ix_1 y_2 + iy_1 x_2 + i^2 y_1 y_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1)$$

4. **Complex Conjugate:** The **conjugate** of $z = x + iy$, denoted $\bar{z}$ or z^* , is obtained by reversing the sign of the imaginary part:

$$\bar{z} = x - iy$$

Conjugate Properties

:

$$\overline{z_1 \pm z_2} = \bar{z}_1 \pm \bar{z}_2, \quad \overline{z_1 z_2} = \bar{z}_1 \bar{z}_2, \quad \overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}$$

$$z + \bar{z} = 2x = 2\operatorname{Re}(z) \implies \operatorname{Re}(z) = \frac{z + \bar{z}}{2}$$

$$z - \bar{z} = 2iy = 2i\operatorname{Im}(z) \implies \operatorname{Im}(z) = \frac{z - \bar{z}}{2i}$$

$$z\bar{z} = (x + iy)(x - iy) = x^2 - i^2 y^2 = x^2 + y^2 \geq 0 \quad (\text{Always a real, non-negative quantity!})$$

5. **Division:** To divide by a complex number $z_2 \neq 0$, multiply both numerator and denominator by the conjugate $\bar{z}_2$ to realify the denominator:

$$\frac{z_1}{z_2} = \frac{x_1 + iy_1}{x_2 + iy_2} \cdot \frac{x_2 - iy_2}{x_2 - iy_2} = \frac{(x_1 x_2 + y_1 y_2) + i(y_1 x_2 - x_1 y_2)}{x_2^2 + y_2^2} = \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} + i \frac{y_1 x_2 - x_1 y_2}{x_2^2 + y_2^2}$$

D. Modulus & The Triangle Inequality

The **modulus** (or absolute value) of $z = x + iy$ is the Euclidean distance from the origin $(0, 0)$ to the point (x, y) in the complex plane:

$$|z| = \sqrt{x^2 + y^2} = \sqrt{z\bar{z}}$$

Modulus Properties

:

- $|z| \geq 0$, and $|z| = 0 \iff z = 0$
- $|z_1 z_2| = |z_1| |z_2|$
- $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$
- $|\bar{z}| = |z| = |-z|$

The Triangle Inequalities (Theorem 17.1.1): For any complex numbers z_1, z_2 :

$$|z_1 + z_2| \leq |z_1| + |z_2| \quad (\text{Standard Triangle Inequality})$$

$$|z_1 - z_2| \geq ||z_1| - |z_2|| \quad (\text{Reverse Triangle Inequality})$$

2.2 Polar & Exponential Forms (Zill §17.1)

A. Polar Coordinates in the Complex Plane

Let (r, θ) be polar coordinates of the point (x, y) representing z :

$$x = r \cos \theta, \quad y = r \sin \theta$$

where $r = |z| = \sqrt{x^2 + y^2} \geq 0$. Substituting into $z = x + iy$:

$$z = r(\cos \theta + i \sin \theta)$$

$$\begin{aligned}
\text{Addition:} \quad z_1 + z_2 &= (x_1 + iy_1) + (x_2 + iy_2) = (x_1 + x_2) + i(y_1 + y_2) \\
\text{Subtraction:} \quad z_1 - z_2 &= (x_1 + iy_1) - (x_2 + iy_2) = (x_1 - x_2) + i(y_1 - y_2) \\
\text{Multiplication:} \quad z_1 \cdot z_2 &= (x_1 + iy_1)(x_2 + iy_2) \\
&= x_1x_2 - y_1y_2 + i(y_1x_2 + x_1y_2) \\
\text{Division:} \quad \frac{z_1}{z_2} &= \frac{x_1 + iy_1}{x_2 + iy_2} \\
&= \frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2} + i \frac{y_1x_2 - x_1y_2}{x_2^2 + y_2^2}
\end{aligned}$$

Figure 17.1.1: Geometric representation of $z = x + iy$ in the Argand plane showing Cartesian coordinates (x, y) , radial distance (modulus) r , and angle (argument) θ .

B. The Argument $\arg(z)$ and Principal Argument $\text{Arg}(z)$

The angle θ measured counterclockwise from the positive real axis is called an **argument** of z , denoted $\theta = \arg(z)$. Because $\cos \theta$ and $\sin \theta$ are 2π -periodic:

$$\arg(z) = \theta + 2k\pi, \quad k \in \mathbb{Z}$$

To eliminate ambiguity, we define the **Principal Argument**, denoted $\text{Arg}(z)$ with a capital A , to be the unique angle confined to the half-open interval:

$$-\pi < \text{Arg}(z) \leq \pi$$

The 4-Quadrant Algorithm for $\text{Arg}(z)$:

$$\arg(x + iy) = \begin{cases} \arctan(y/x), & x > 0 \quad (\text{Quadrants I and IV}) \\ \arctan(y/x) + \pi, & x < 0, y \geq 0 \quad (\text{Quadrant II}) \\ \arctan(y/x) - \pi, & x < 0, y < 0 \quad (\text{Quadrant III}) \end{cases}$$

C. Euler's Formula & The Exponential Form

Euler's Formula: For any real angle θ :

$$e^{i\theta} = \cos \theta + i \sin \theta$$

This gives the compact **Exponential Form** of a complex number:

$$z = re^{i\theta}$$

D. Geometric Multiplication and Division

Let $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$:

1. Multiplication (Moduli Multiply, Arguments Add):

$$z_1 z_2 = (r_1 e^{i\theta_1})(r_2 e^{i\theta_2}) = (r_1 r_2) e^{i(\theta_1 + \theta_2)} = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

Geometry

: Multiplying by z_2 dilates z_1 by a factor of r_2 and rotates it counterclockwise by angle θ_2 .

2. Division (Moduli Divide, Arguments Subtract):

$$\frac{z_1}{z_2} = \frac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} = \left(\frac{r_1}{r_2} \right) e^{i(\theta_1 - \theta_2)} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$$

3. Powers and Roots: De Moivre's Formula (Zill §17.2)

3.1 Integer Powers & De Moivre's Theorem

A. Statement and Proof of De Moivre's Formula

If $z = r(\cos \theta + i \sin \theta) = re^{i\theta}$, then for any positive integer n :

$$z^n = (re^{i\theta})^n = r^n e^{in\theta} = r^n [\cos(n\theta) + i \sin(n\theta)]$$

Setting $r = 1$ yields **De Moivre's Formula**:

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta) \quad \text{for all } n \in \mathbb{Z}$$

Extension to Negative Integers

: If $n = -m$ where $m \in \mathbb{Z}^+$:

$$(\cos \theta + i \sin \theta)^{-m} = \frac{1}{(\cos \theta + i \sin \theta)^m} = \frac{1}{\cos(m\theta) + i \sin(m\theta)} = \cos(m\theta) - i \sin(m\theta) = \cos(-m\theta) + i \sin(-m\theta)$$

Thus De Moivre's formula holds for **all integers** $n \in \mathbb{Z}$.

B. Application: Derivation of Trigonometric Identities

We can derive multiple-angle trigonometric identities without messy algebra by expanding De Moivre's formula using the Binomial Theorem.

•

Derivation of $\cos(3\theta)$ and $\sin(3\theta)$

:

$$\begin{aligned} \cos(3\theta) + i \sin(3\theta) &= (\cos \theta + i \sin \theta)^3 \\ &= \cos^3 \theta + 3 \cos^2 \theta (i \sin \theta) + 3 \cos \theta (i \sin \theta)^2 + (i \sin \theta)^3 \\ &= \cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta \end{aligned}$$

Group real and imaginary terms:

$$\cos(3\theta) + i \sin(3\theta) = (\cos^3 \theta - 3 \cos \theta \sin^2 \theta) + i(3 \cos^2 \theta \sin \theta - \sin^3 \theta)$$

Equating real parts:

$$\cos(3\theta) = \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta) = 4 \cos^3 \theta - 3 \cos \theta$$

Equating imaginary parts:

$$\sin(3\theta) = 3(1 - \sin^2 \theta) \sin \theta - \sin^3 \theta = 3 \sin \theta - 4 \sin^3 \theta$$

3.2 Roots of Complex Numbers (n -th Roots)

A. Formulation of the Root Equation

Given a non-zero complex constant $z_0 = r_0 e^{i\theta_0}$, we wish to find all complex numbers $w = \rho e^{i\phi}$ satisfying:

$$w^n = z_0$$

Substituting exponential forms:

$$(\rho e^{i\phi})^n = \rho^n e^{in\phi} = r_0 e^{i\theta_0}$$

Because $e^{i\theta}$ is 2π -periodic, the exponential factors match whenever the exponents differ by an integer multiple of $2\pi i$:

1. Modulus Equality:

$$\rho^n = r_0 \implies \rho = r_0^{1/n} = \sqrt[n]{r_0} \quad (\text{Unique positive real } n\text{-th root})$$

2. Argument Equality:

$$n\phi = \theta_0 + 2k\pi \implies \phi_k = \frac{\theta_0 + 2k\pi}{n}, \quad k \in \mathbb{Z}$$

As k takes values $0, 1, 2, \dots, n-1$, we generate **exactly n distinct angles** ϕ_k . For $k = n$, the angle becomes $\frac{\theta_0 + 2n\pi}{n} = \frac{\theta_0}{n} + 2\pi$, which returns to the angle for $k = 0$.

B. The Master n -th Root Formula (Theorem 17.2.1)

The n distinct n -th roots of $z_0 = r_0 [\cos \theta_0 + i \sin \theta_0]$ are given by:

$$w_k = r_0^{1/n} \left[\cos \left(\frac{\theta_0 + 2k\pi}{n} \right) + i \sin \left(\frac{\theta_0 + 2k\pi}{n} \right) \right], \quad k = 0, 1, 2, \dots, n-1$$

Or in exponential shorthand:

$$w_k = r_0^{1/n} \exp \left(i \frac{\theta_0 + 2k\pi}{n} \right), \quad k = 0, 1, 2, \dots, n-1$$

C. Geometric Distribution of Roots

The roots possess a magnificent geometric property:

- All n roots lie on a **single circle centered at the origin** of radius:

$$R = r_0^{1/n}$$

- The roots are **spaced at equal angular increments** of:

$$\Delta\phi = \frac{2\pi}{n} \text{ radians} = \frac{360^\circ}{n}$$

- The roots form the **vertices of a regular n -sided polygon** inscribed within the circle!

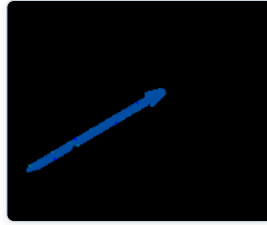


Figure 17.2.1: The n distinct roots of a complex number evenly spaced around a circle of radius $r^{1/n}$ in the complex plane, forming a regular polygon.

D. The n -th Roots of Unity

A fundamental special case is solving $w^n = 1$. Here $r_0 = 1$ and $\theta_0 = 0$. The n roots of unity are:

$$\omega_k = \exp\left(i\frac{2k\pi}{n}\right) = \left[\exp\left(i\frac{2\pi}{n}\right)\right]^k = \omega_n^k, \quad k = 0, 1, 2, \dots, n-1$$

where $\omega_n = e^{i2\pi/n}$ is called the **primitive n -th root of unity**.

Sum of Roots Property

: The sum of all n -th roots of unity is identically zero:

$$\sum_{k=0}^{n-1} \omega_n^k = 1 + \omega_n + \omega_n^2 + \dots + \omega_n^{n-1} = \frac{1 - \omega_n^n}{1 - \omega_n} = \frac{1 - 1}{1 - \omega_n} = 0$$

4. Comprehensive Step-by-Step Problem Walkthroughs

4.1 Problem 1: Rectangular Conversion, Polar Arithmetic & Powers

Problem Statement: Given $z_1 = -1 + i\sqrt{3}$ and $z_2 = \sqrt{2} - i\sqrt{2}$:

1. Express z_1 and z_2 in exponential polar form $re^{i\theta}$ using principal arguments.
2. Calculate the exact rectangular forms of $z_1 z_2$ and $\frac{z_1}{z_2}$ using polar multiplication/division.
3. Compute $(z_1)^9$ using De Moivre's Formula.

Step 1: Convert z_1 to Polar Form

- Modulus:

$$r_1 = |z_1| = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{1+3} = \sqrt{4} = 2$$

- Argument: $x = -1 < 0$ and $y = \sqrt{3} > 0 \implies$ Quadrant II:

$$\text{Arg}(z_1) = \arctan\left(\frac{\sqrt{3}}{-1}\right) + \pi = -\frac{\pi}{3} + \pi = \frac{2\pi}{3}$$

- Exponential Form:

$$z_1 = 2e^{i2\pi/3}$$

Step 2: Convert z_2 to Polar Form

- Modulus:

$$r_2 = |z_2| = \sqrt{(\sqrt{2})^2 + (-\sqrt{2})^2} = \sqrt{2+2} = \sqrt{4} = 2$$

- Argument: $x = \sqrt{2} > 0$ and $y = -\sqrt{2} < 0 \implies$ Quadrant IV:

$$\text{Arg}(z_2) = \arctan\left(\frac{-\sqrt{2}}{\sqrt{2}}\right) = \arctan(-1) = -\frac{\pi}{4}$$

- Exponential Form:

$$z_2 = 2e^{-i\pi/4}$$

Step 3: Compute $z_1 z_2$

$$z_1 z_2 = (r_1 r_2) e^{i(\theta_1 + \theta_2)} = (2 \cdot 2) \exp \left(i \left[\frac{2\pi}{3} - \frac{\pi}{4} \right] \right) = 4 \exp \left(i \frac{8\pi - 3\pi}{12} \right) = 4e^{i5\pi/12}$$

In exact rectangular form:

$$z_1 z_2 = 4 \left(\cos \frac{5\pi}{12} + i \sin \frac{5\pi}{12} \right)$$

Using angle sum identities: $\frac{5\pi}{12} = \frac{\pi}{6} + \frac{\pi}{4}$:

- $\cos \frac{5\pi}{12} = \cos \frac{\pi}{6} \cos \frac{\pi}{4} - \sin \frac{\pi}{6} \sin \frac{\pi}{4} = \left(\frac{\sqrt{3}}{2} \right) \left(\frac{\sqrt{2}}{2} \right) - \left(\frac{1}{2} \right) \left(\frac{\sqrt{2}}{2} \right) = \frac{\sqrt{6} - \sqrt{2}}{4}$
- $\sin \frac{5\pi}{12} = \sin \frac{\pi}{6} \cos \frac{\pi}{4} + \cos \frac{\pi}{6} \sin \frac{\pi}{4} = \left(\frac{1}{2} \right) \left(\frac{\sqrt{2}}{2} \right) + \left(\frac{\sqrt{3}}{2} \right) \left(\frac{\sqrt{2}}{2} \right) = \frac{\sqrt{2} + \sqrt{6}}{4}$ Multiplying by 4:

$$z_1 z_2 = (\sqrt{6} - \sqrt{2}) + i(\sqrt{6} + \sqrt{2})$$

Step 4: Compute z_1 / z_2

$$\frac{z_1}{z_2} = \left(\frac{r_1}{r_2} \right) e^{i(\theta_1 - \theta_2)} = \left(\frac{2}{2} \right) \exp \left(i \left[\frac{2\pi}{3} - \left(-\frac{\pi}{4} \right) \right] \right) = 1 \cdot e^{i11\pi/12}$$

Using $\frac{11\pi}{12} = \pi - \frac{\pi}{12}$:

$$\begin{aligned} \cos \frac{11\pi}{12} &= -\frac{\sqrt{6} + \sqrt{2}}{4}, & \sin \frac{11\pi}{12} &= \frac{\sqrt{6} - \sqrt{2}}{4} \\ \frac{z_1}{z_2} &= -\frac{\sqrt{6} + \sqrt{2}}{4} + i \frac{\sqrt{6} - \sqrt{2}}{4} \end{aligned}$$

Step 5: Compute $(z_1)^9$ via De Moivre's Theorem

$$(z_1)^9 = (2e^{i2\pi/3})^9 = 2^9 e^{i(9 \cdot 2\pi/3)} = 512e^{i6\pi}$$

Since 6π is an even multiple of π , $e^{i6\pi} = \cos(6\pi) + i \sin(6\pi) = 1 + 0i = 1$:

$$(z_1)^9 = 512(1) = 512$$

4.2 Problem 2: Finding All Complex Roots ($w^4 = -8 - 8\sqrt{3}i$)

Problem Statement: Find all 4 roots of the complex equation $w^4 = -8 - 8\sqrt{3}i$. Express each root in exact rectangular form $a + ib$ and sketch their distribution in the complex plane.

Step 1: Express $z_0 = -8 - 8\sqrt{3}i$ in Polar Form

- Modulus r_0 :

$$r_0 = \sqrt{(-8)^2 + (-8\sqrt{3})^2} = \sqrt{64 + 64(3)} = \sqrt{64(1+3)} = \sqrt{256} = 16$$

- Argument θ_0 : Both $x = -8 < 0$ and $y = -8\sqrt{3} < 0 \implies$ **Quadrant III**:

$$\text{Arg}(z_0) = \arctan \left(\frac{-8\sqrt{3}}{-8} \right) - \pi = \arctan(\sqrt{3}) - \pi = \frac{\pi}{3} - \pi = -\frac{2\pi}{3}$$

(Alternatively, using positive angle: $\theta_0 = \frac{4\pi}{3}$)

$$z_0 = 16 \exp \left(-i \frac{2\pi}{3} \right) = 16 \exp \left(i \frac{4\pi}{3} \right)$$

Step 2: Set Up the n -th Root Equation for $n = 4$

The root modulus is:

$$\rho = r_0^{1/4} = 16^{1/4} = 2$$

The root angles are:

$$\phi_k = \frac{\theta_0 + 2k\pi}{4} = \frac{\frac{4\pi}{3} + 2k\pi}{4} = \frac{\pi}{3} + \frac{k\pi}{2}, \quad k = 0, 1, 2, 3$$

The angular increment between adjacent roots is:

$$\Delta\phi = \frac{2\pi}{4} = \frac{\pi}{2} = 90^\circ$$

Step 3: Evaluate Each of the 4 Roots

- **Root $k = 0$:**

$$\phi_0 = \frac{\pi}{3}$$

$$w_0 = 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = 2 \left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right) = 1 + i\sqrt{3}$$

- **Root $k = 1$:**

$$\phi_1 = \frac{\pi}{3} + \frac{\pi}{2} = \frac{5\pi}{6}$$

$$w_1 = 2 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right) = 2 \left(-\frac{\sqrt{3}}{2} + i \frac{1}{2} \right) = -\sqrt{3} + i$$

- **Root $k = 2$:**

$$\phi_2 = \frac{\pi}{3} + \pi = \frac{4\pi}{3}$$

$$w_2 = 2 \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right) = 2 \left(-\frac{1}{2} - i \frac{\sqrt{3}}{2} \right) = -1 - i\sqrt{3}$$

(Notice that $w_2 = -w_0$, lying directly opposite w_0 across the origin!)

- **Root $k = 3$:**

$$\phi_3 = \frac{\pi}{3} + \frac{3\pi}{2} = \frac{11\pi}{6}$$

$$w_3 = 2 \left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6} \right) = 2 \left(\frac{\sqrt{3}}{2} - i \frac{1}{2} \right) = \sqrt{3} - i$$

(Notice that $w_3 = -w_1$!)

Step 4: Geometric Verification & Polygon Check

- Radius of root circle: $R = 2$.
- The four roots:

$$w_0 = 1 + i\sqrt{3}, \quad w_1 = -\sqrt{3} + i, \quad w_2 = -1 - i\sqrt{3}, \quad w_3 = \sqrt{3} - i$$

- They form the **four vertices of a square centered at the origin**, each rotated by 90° relative to the next!

4.3 Problem 3: Factoring Polynomials with Complex Conjugate Roots

Problem Statement: Solve $z^4 + 16 = 0$ in $\mathbb{C}$, and use the roots to factor $z^4 + 16$ into the product of two real irreducible quadratic factors.

Step 1: Solve for the Roots of $z^4 = -16$

- Modulus: $r_0 = |-16| = 16 \implies \rho = 16^{1/4} = 2$.
- Argument: $\theta_0 = \pi$.
- Angles for $k = 0, 1, 2, 3$:

$$\phi_k = \frac{\pi + 2k\pi}{4} = \frac{\pi}{4} + \frac{k\pi}{2}$$

$$\circ k = 0: \phi_0 = \frac{\pi}{4} \implies z_0 = 2 \left(\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2} \right) = \sqrt{2} + i\sqrt{2}$$

- $k = 1: \phi_1 = \frac{3\pi}{4} \implies z_1 = 2 \left(-\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} \right) = -\sqrt{2} + i\sqrt{2}$
- $k = 2: \phi_2 = \frac{5\pi}{4} \implies z_2 = 2 \left(-\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} \right) = -\sqrt{2} - i\sqrt{2}$
- $k = 3: \phi_3 = \frac{7\pi}{4} \implies z_3 = 2 \left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} \right) = \sqrt{2} - i\sqrt{2}$

Step 2: Pair Conjugate Roots

Notice the conjugate pairs:

- Pair 1: $z_0 = \sqrt{2} + i\sqrt{2}$ and $z_3 = \bar{z}_0 = \sqrt{2} - i\sqrt{2}$.
- Pair 2: $z_1 = -\sqrt{2} + i\sqrt{2}$ and $z_2 = \bar{z}_1 = -\sqrt{2} - i\sqrt{2}$.

Step 3: Multiply Conjugate Factors to Form Real Quadratics

- **First quadratic factor from Pair 1:**

$$(z - z_0)(z - \bar{z}_0) = z^2 - (z_0 + \bar{z}_0)z + z_0\bar{z}_0$$

$$z_0 + \bar{z}_0 = 2 \operatorname{Re}(z_0) = 2\sqrt{2}$$

$$z_0\bar{z}_0 = |z_0|^2 = (\sqrt{2})^2 + (\sqrt{2})^2 = 2 + 2 = 4$$

$$(z - z_0)(z - \bar{z}_0) = z^2 - 2\sqrt{2}z + 4$$

- **Second quadratic factor from Pair 2:**

$$(z - z_1)(z - \bar{z}_1) = z^2 - (z_1 + \bar{z}_1)z + z_1\bar{z}_1$$

$$z_1 + \bar{z}_1 = 2 \operatorname{Re}(z_1) = 2(-\sqrt{2}) = -2\sqrt{2}$$

$$z_1\bar{z}_1 = |z_1|^2 = (-\sqrt{2})^2 + (\sqrt{2})^2 = 4$$

$$(z - z_1)(z - \bar{z}_1) = z^2 + 2\sqrt{2}z + 4$$

Step 4: Final Real Factorization

$$z^4 + 16 = (z^2 - 2\sqrt{2}z + 4)(z^2 + 2\sqrt{2}z + 4)$$

(This exact factorization is required when performing partial fractions on $\frac{1}{s^4+16}$ in Laplace transforms!)

5. Critical Concordia Exam Traps & Red Flags

- **⚠ Trap 1: Quadrant Failure in $\arctan(y/x)$:** Calculators always return $\arctan(y/x) \in (-\pi/2, \pi/2)$. If $z = -1 - i$, $y/x = 1$, and calculator gives $\pi/4$. But z is in Quadrant III! Writing $\theta = \pi/4$ will result in a zero on the problem. You must adjust: $\operatorname{Arg}(z) = \frac{\pi}{4} - \pi = -\frac{3\pi}{4}$.
- **⚠ Trap 2: Losing Roots by Forgetting $2k\pi$:** When solving $w^n = z_0$, students often write $\phi = \frac{\theta_0}{n}$ and stop after finding only one root. You **must** add $\frac{2k\pi}{n}$ for $k = 0, 1, \dots, n-1$ to extract all n distinct roots!
- **⚠ Trap 3: Taking the n -th Root of Modulus Incorrectly:** In $w^4 = 16e^{i\pi}$, the modulus of each root is $\sqrt[4]{16} = 2$. Students sometimes mistakenly divide the modulus by n ($16/4 = 4$). You take the n -th **root** of the modulus, but you **divide** the argument by n !
- **⚠ Trap 4: Missing Minus Sign in Conjugate Division:** When dividing $\frac{z_1}{z_2}$, always write the denominator as $x_2^2 + y_2^2$ (a **sum** of squares). Because $i^2 = -1$, $(x_2 + iy_2)(x_2 - iy_2) = x_2^2 - i^2y_2^2 = x_2^2 + y_2^2$. Never subtract squares in the denominator!